21-701: Discrete Mathematics

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Chapter 1

Graphs and Colorings

Much of this course will be about graphs, and a surprising amount of what we want to know about a graph can be phrased as a question about coloring it. We begin with the vocabulary, and then settle the first such question completely: which graphs admit a coloring with only two colors. We then ask the same question of hypergraphs, where it becomes considerably harder—even pinning down how many edges a 3-uniform hypergraph needs before two colors are no longer enough takes real work.

Notation.

- $[k]$ will denote $\{1, \ldots, k\}$.
- $\sim$ will denote the adjacency relation on graph vertices.
- K_n will denote the complete graph on n vertices.

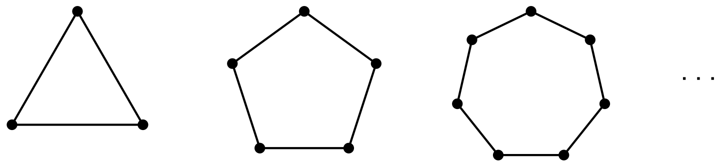
1.1 Colorings

Definition 1.1.1 (Coloring)**.** Given a graph $G = (V, E)$, a k-coloring of G is a map $c : V \to [k]$ such that $u \sim v \implies c(u) \neq c(v)$.

1.1.1 2-Colorings of Graphs

Let's start with a motivating question: which graphs can be 2-colored?

Any odd cycle graph can't be 2-colored. That is, any graph that looks like



Let's study when odd cycles exist.

Lemma 1.1.2. *If G has an odd closed walk, then G has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbor, return).

Proof. Suppose G has an odd closed walk. Let W be such a walk of minimal length. Write $W = (x_1, \ldots, x_i, \ldots, x_j, \ldots, x_\ell)$, with $x_\ell = x_1$. Suppose W is not a cycle. Then there must be some x_i, x_j such that $x_i = x_j$ even though $i \neq j$, ie, there must be some repetition. Then, $(x_1, \ldots, x_i, x_{j+1}, \ldots, x_\ell)$ and $(x_i, x_{i+1}, \ldots, x_j)$ are both shorter walks than W. Moreover, one of them must be of odd length. This contradicts minimality of the length of W. So W must be a cycle. □

Here's a cool result.

Theorem 1.1.3. *A graph $G = (V, E)$ can be 2-colored if and only if G has no odd cycle.*

Proof.

($\implies$) We've already seen that odd cycles can't be 2-colored. So if G can be 2-colored, it can't have an odd cycle. We've essentially seen this from the examples above.

($\impliedby$) We may assume G is connected, since otherwise we can color each component separately. Fix some $v_0 \in V$. Define $c : V \to \{1, 2\}$ by

$$c(u) = \begin{cases} 2 & \text{if } \operatorname{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \operatorname{dist}(v_0, u) \text{ is even} \end{cases}$$

where $\mathrm{dist}(\cdot,\cdot)$ denotes the shortest number of edges from one vertex to another.

We claim that this c is a valid 2-coloring.

Indeed, suppose it is not. Then, there must be some $u, v \in V$ such that $u \sim v$ and $c(u) = c(v)$. Now let's consider the two possible cases on the color.

Suppose $c(u) = c(v) = 1$ for adjacent u, v. Then, there is a path of even length between v and v_0 and another path of even length between u and v_0. Concatenating the path from v to v_0 with the one from v_0 to u, and closing up with the edge uv, we get a closed walk of length even + even +1. This is odd, so by Theorem 1.1.2, G has an odd cycle.

The case $c(u) = c(v) = 2$ is identical: the two paths are now of odd length, so the closed walk they form with the edge uv has length odd + odd +1, which is odd again.

Either way, G has an odd cycle, contrary to hypothesis. So c must be a valid 2-coloring.

□

1.1.2 Colorings of Hypergraphs

Coloring is not really a question about graphs. All the definition above asks of G is that we know which pairs of vertices may not share a color, and nothing stops us forbidding larger sets instead.

Definition 1.1.4 (Hypergraph). Given a vertex set V, a **hypergraph** is a collection H of subsets of V. The sets in H are called **hyperedges**.

Definition 1.1.5 (Uniformity). A hypergraph H is k-**uniform** if $|f| = k$ for all $f \in H$.

The following is thus obvious.

Proposition 1.1.6. *A graph is a 2-uniform hypergraph.*

Definition 1.1.7. A proper coloring of a hypergraph H is an assignment $c : V \to [r]$ such that $\forall f \in H$, $|c(f)| \geq 2$.

We say "no edge is monochromatic".

We can ask ourselves the following question: **"Which 3-uniform hypergraphs are 2-colorable?**

For each k, define $m(k)$ to be the smallest m such that $\exists$ some k-uniform hypergraph with m edges that is not 2-colorable. We can show that $m(2) = 3$. But what is $m(3)$? That is, how many 3-edges do I need to prevent 2-colorability?

Let H be a 3-uniform hypergraph that's not 2-colorable. Assume, further, that H has at most 6 edges (each being a subset of size 3).

First, we'll show that we can reduce, without loss of generality, to the case where H has exactly 6 vertices. Indeed, if H has vertices x, y that are not in any common edge, then I can define H' by "merging" x and y. So H' has one fewer vertex than H. This H' cannot be 2-colorable: if it were, then that coloring would extend to H by using the color of the merged vertex at both x and y, which would make H 2-colorable. And when H has more than 6 vertices, such a pair always exists.

An edge has 3 vertices, so it covers the $\binom{3}{2} = 3$ pairs among them, and H has at most 6 edges, so at most

$$6 \cdot \binom{3}{2} = 18$$

pairs are covered in total. But if H has $n \geq 7$ vertices, there are $\binom{n}{2} \geq \binom{7}{2} = 21$ pairs to cover. So some pair is left uncovered, and merging it brings us one vertex closer to 6.

So now assume that H has exactly 6 edges and 6 vertices and is not 2-colorable.

1.1.3 Ramsey Numbers

For this subsection, let's step back into "graph land" from "hypergraph land".

> **Definition 1.1.8** (Ramsey Number). The kth Ramsey number, denoted $R(k)$, is the smallest $n \in \mathbb{N}$ such that any 2-colouring of edges of K_n gives a monochromatic K_k.

One immediate result we can see is that $R(3) \geq 6$, which we can see simply by drawing a picture of K_5 and studying what happens if we move our attention to K_6.

We can actually prove that $R(3) = 6$, but this is rather nontrivial.

Other known Ramsey numbers are $R(2) = 2$ and $R(4) = 18$. The rest are unknown, though we do have bounds for them. The best (or close to the best) known bounds are

$$2^{\frac{k}{2}} \leq R(k) \leq 3.8^k$$

Theorem 1.1.9. *$m(k) \geq 2^{k-1}$. In other words, any k-uniform hypergraph with $< 2^{k-1}$ edges is 2-colourable.*

Proof. Given such a hypergraph, randomly colour the vertices red and blue, with equal probability (1/2) for each. For each edge f, let E_f be the event that f is monochromatic. Then,

$$\Pr(E_f) = \frac{2}{2^k}$$

Taking a simple union bound,

$$\Pr\left(\bigcup_{f \in H} E_f\right) \leq \frac{|H|}{2^{k-1}} < 1$$

where we interpret $\bigcup_{f \in H} E_f$ as the event that there is a monochromatic edge. As the probability is less than 1, there must be at least one colouring with no monochromatic edge. $\square$

Note that the above argument can also be written combinatorially instead of probabilistically. In this instance, the probabilistic language essentially conveys the same information as combinatorial language would. But this may not always be true, so it's worth being careful about this.

Exercise 1.1.10. Prove that $m(k) = O\left(k^2 \cdot 2^k\right)$

Here's a fun fact: a "maker-breaker" game is a game where the "maker" wins if they are able to achieve an objective and the "breaker" wins if they are able to stop the maker from winning. So if you set up a "maker-breaker" version of tic-tac-toe, where only the crosses can win by making three in a row and the noughts can only win by stopping the crosses, the crosses will always win (assuming the crosses play first). This is, of course, different from tic-tac-toe, which, as our own studies from when we were a kid have shown, can always be drawn.

Theorem 1.1.11 (Erdős-Selfridge). *If H is a k-uniform hypergraph with m edges, and $m < 2^{k-1}$, then breaker wins (if breaker goes second).*

In the language of hypergraphs, the game is 2-colouring; maker winning is equivalent to the existence of a colouring with a monochromatic edge; and breaker winning is equivalent to the inexistence of such edge.

Proof. Let Δ be the maximum degree of H, that is,

$$\Delta = \max_{v \in H} |\{f \subseteq H \mid v \in f\}|$$

We'll prove that if $m + \Delta < 2^k$, then breaker will win. Then, since $\Delta \leq m$, it will follow that if $m < 2^{k-1}$, then $m + \Delta \leq 2m < 2^k$.

At the start of move i, we will conceptualise the danger of edge $f \in H$ by

$$\phi_i(f) = \begin{cases} 2^{\text{number of maker moves in } f} & \text{if there are no breaker moves in } f \\ 0 & \text{if there is a breaker move in } f \end{cases}$$

$$\Phi_i(f) = \sum_{f \in H} \phi_i(f)$$

Consider the strategy where each time maker plays y on move i, breaker plays x to minimise the sum

$$\sum_{f \ni x} \phi_i(f) + \sum_{f \ni y} \phi_i(f)$$

One can show that with this strategy, the danger Φ_i never exceeds $m + \Delta$. □

Let's consider higher-dimensional maker-breaker tic-tac-toe. Turns out that $8 \times 8 \times 8$ maker-breaker tic-tac-toe is breaker-winnable: we can show that $m = \frac{10^3 - 8^3}{2} = 244$ and $\Delta = 7$, meaning $m + \Delta = 251 < 2^k = 256$.

We won't say any more about tic-tac-toe in this course, but it was a fun aside.

Bibliography

These lecture notes are based heavily on the following references:

[1] Nicolas Bourbaki. *Groupes et Algèbres de Lie. Chapitre 1*. 1st ed. Éléments de Mathématique. Édition originale publiée par Hermann, Paris, 1972. Springer Berlin, Heidelberg, 2007. ISBN: 978-3-540-35337-9. DOI: https://doi.org/10.1007/978-3-540-35337-9.

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